

TABLE 2: Details

ITEM	RATIONALE / JUSTIFICATION	COST UGX (£)
	A] CLINICAL EQUIPMENT	(at 4,900 to £)
	1: General wards, clinics & operating theatres	
Multi-parameter patient clinical signs monitors x3	Provision of quality care for emergency cases including critically ill medical, surgical, obstetric & paediatric patients including trauma.. Multi-parameter monitors continuously measure pulse, blood pressure, electrocardiogram, temperature, oxygen saturation and respiratory rate allowing clinical staff to monitor response to treatment. They store data allowing analysis of trends in response to treatment. In covid-suspects they reduce how often staff need to take observations. Currently we have one SaO2 monitor in our Accident & Emergency area yet often need more.	13.5m (£2,755)
Oxygen concentrators x2 & portable x1	These machines extract oxygen from room air to then provide support to patients. This may be life saving. Currently we have one static oxygen concentrator in A&E. Need 2 more so each bed has its supply, ideally 6 litre/minute models. The portable concentrator supports safe transfer of oxygen-dependent patients to x-ray, theatre or wards or other institutions where needed.	10m (£2,040)
Resuscitation equipment	All emergency resuscitation equipment immediately available for all ages from neonates to adults. Airway support including laryngoscopes, endotracheal tubes, ambu bags. Circulatory support including intra-osseus and intravenous access. Drugs including inotropes.	3m (£612)
Resuscitation training	Further develop multi-disciplinary training to ensure co-ordinated team work e.g. anaesthetists, lab, pharmacy, medical and nursing teams	0 (in-house)
Trolley beds x3	These are used to stabilise patients in the Accident department and allow movement to wards, x-ray, operating theatres etc. Current trolley beds are getting old and less reliable. Need to be able to reposition patients easily e.g. in resuscitation situations.	6m (£1,224)
Diagnostic sets x2	Instruments used by doctors during examination of patients to help make a diagnosis. Ophthalmoscopes to look at the retina may show raised intra-cranial pressure, Otoscope to check ears may show infections, bleeding etc.	1.9m (£388)
Operating Theatre Instrument sets	Many of our surgical instrument sets for different procedures are decades old with some items becoming unreliable which is unacceptable for safe operating. Some sets are incomplete requiring surgeons to improvise with alternative instruments which is not ideal	26.6m (£5,428) <i>See Appendix 1</i>
Sterilising drums	4 replacement sterilising drums needed for Autoclave use.	12m (£2,500)
Autoclave	Autoclave sterilise by the use of steam under high pressure to avoid any risk of infection being transmitted. Kisiizi has two old autoclaves that are becoming less reliable. If we cannot autoclave then we cannot operate safely in maternity or general theatres. A new main autoclave will help us ensure sterile theatre & maternity instruments, drapes and equipment are always available.	12m (£2,449)
Operating theatre ceiling lights x2	We have successfully installed these lights in two of our four general opearating theatres and it has significantly improved quality. We now need to install the same in the two remaining theatres to replace old second-hand lights that are becoming unreliable and potentially unsafe.	44m (£8,980)
Nursing equipment	Surgical ward e.g. metal stainless steel kidney dishes, forceps, scissors etc. for dressings of wounds, burns etc. Many wards have aging equipment that needs to be replaced. (see Appendix 2)	8.3m (£1,694)
Personal Protective E.	Equipment to protect staff from infections including Covid-19. Masks, Gloves, Aprons, hand-rub, Hypochlorite solution, gowns etc.	25m (£5,102)

Spirometry equipment	Allows the physicians to perform objective assessment of respiratory problems with accurate diagnosis avoiding deterioration. Will help large numbers of patients and avoid some being started on unnecessary treatments. 12 – 20 patients a month are <u>admitted</u> with exacerbations of chronic lung diseases.	12m (£2,448)
Dental chair complete including suction unit	Dental suction system collects blood, saliva and other debris generated during dental procedures and avoids the risk of aspiration/choking in a patient who unexpectedly begins vomiting or bleeding. Use of suction offers a significant reassurance to patients, they feel less anxious about the treatment. Improves the working environment for dental personnel which shortens the time spent per patient and improves the quality of dental restorations that are very sensitive to saliva.	18m (£3,673)
Dental Autoclave 18 litres x1	Autoclaves sterilise instruments used in dentistry to avoid infections. Currently we have an aging slow cycle-time autoclave that is becoming risky as the door closure system is wearing down with a future risk of it blowing open as the system uses high pressure.	7.3m (£1,489)
Dental x-ray x1	Improves diagnostic accuracy and monitoring of progress providing better quality outcomes. Our old apparatus is becoming unreliable.	8.6m (£1,755)
Spare parts	Items such as batteries, chargers, keypads, bulbs needed to repair some medical and laboratory equipment	7m (£1,428)
	2: Laboratory Equipment	
Blood culture service	Many of our patients have bacterial infections that may be life-threatening. This includes children. Accurate diagnosis of the organism causing septicaemia and other infections is important and the ability to test antibiotic sensitivities allows correct choice of antibiotics. Patients who are on the wrong treatment may continue to deteriorate. These cultures require a machine to perform and read the cultures plus bottles for the samples. It needs to be standardised to avoid cross-contamination to ensure accurate results.	93m (£18,979)
Full blood count analyser	This Analyser includes 5 parameters of white blood cell count for improved diagnosis guidance for clinicians compared to the 3 parameter old machine.	45m (£9,183)
Microscope	For diagnostic use e.g. diagnosis of malaria, parasites, bacterial infections etc. We have inadequate capacity for the demand which can cause delays in patient care.	8m (£1,632)
	3: Imaging Equipment	
Portable ultrasound scanner x2	Needed to perform basic obstetric scans that will identify the number of babies, which way the baby is facing, the site of the placenta etc. and help to assess the gestation in 1] maternity for women arriving in labour; 2] for the imaging department to be used in various situations as needed around the hospital including ultrasound guided biopsy in theatre, cranial ultrasound in the special care baby unit etc.	30m (£6,122)
Mobile x-ray / C-arm x-ray	Our current static x-ray machine means patients in theatre or high-dependency units have to be moved to the imaging department. If the x-ray is to check position after manipulation of a fracture, the patient has to be woken up from anaesthesia with the chance of needing a second anaesthetic if the alignment is unsatisfactory.	60m (£12,245)
CT scanner	We currently have to send patients in an ambulance to Kabale, about 90 minutes away, for CT scans, mainly of the brain for conditions such as head injury, coma, stroke etc. to confirm the precise diagnosis and so give the correct treatments. It is also used in a range of other situations such as potential cancers. Some patients cannot afford the scans and the transport costs. The equipment is expensive and would need a small building specifically to house it, air-conditioning, power surge protection and stabilizer etc. but if it was obtained it would allow faster scans and responses and would benefit patients in the region who would be able to come here from other health units for reliable investigation. We have a member of our staff who has had imaging training and can read and report the scans.	995m (£203k)
Mains cable for CT	70m from hydro-electricity generator house	4.2m (£857)
	4: Maternity equipment	
Sonicaids x5	These instruments help to monitor foetal (baby) heart rates and rhythms in pregnancy and before delivery to help staff decide if interventions are needed in antenatal clinic, maternity and maternity theatre.	2.56m (£522)
Delivery packs x 5	Instruments and equipment needed for safe maternity delivery. Needed to replace old equipment. We need enough sets to ensure there	1.5m (£306)

	is always a sterile pack ready for any unexpected admissions who may deliver very quickly.	
Beds x20	Current beds are old and some are failing and need replacement for safety reasons and to ensure that good hygiene may be practiced.	25m (£5,102)
	B] SUPPORT INFRASTRUCTURE EQUIPMENT	
Solar water heaters x32	For clinical departments and staff residences will free up about 50kW of electric power for the hydro-electricity grid reducing the risk of power cuts in the dry season so better continuity of clinical services and reducing running costs to the hospital.	121m (£24,693)
Solar back up 10kW	For School of Nursing to avoid power cuts especially in the dry season. Power cuts may cause security risks and interrupt student studies etc. The back up plant should also reduce ongoing running costs and the demand on the main hospital hydro-electricity supply.	95m (£19,387)
Uninterrupted Power Supply (UPS)/surge	UPS continues to supply power for some time to ensure continuity with lab or medical equipment or computers. Surge protectors stop delicate equipment being damaged if the voltage of supply changes suddenly. APC Smart UPS 3000VA for lab (6m), 20 computers (7.5m)	13.5m (£2,755)
Computer network Server	Current server is over 7 years old, many systems depend on it so needs to be replaced as it is becoming unreliable	7m (£1,428)
Desktop computers x2	For Community Health Insurance Scheme: Two current computers are more than 10 years old and slow affecting flow of patients and need to be replaced	4.3m (£877)
Stre@mline development	Integrated patient-centred IT system developed in Kisiizi. Need extra software engineer for 12 months to accelerate development e.g. specialist clinics, maternity module etc. Now in over 20 hospitals in Uganda. Incorporates Community Insurance Scheme.	24m (£4,897)
CCTV	Security protection for staff, reduce risk of theft	6m (£1,224)
	C] STAFFING SUPPORT	(including accommodation)
10 Enrolled nurses	Needed to support Out-Patient services with nurse support for each clinic room. Improve 24-hour cover in different departments including night duty.	60m (£12,244)
Palliative Care nurse	To support new multi-disciplinary service caring for those with terminal diseases offering quality palliative care support.	8m (£1,632)
Second Psychiatric Clinical Officer	Allowing increased community clinics, home visits etc. and better supervision of in-patient service. Will support alcohol & substance abuse services, epilepsy, mhGAP training etc. including the impact of covid.	35m (£7,142)
Child & Adolescent Mental Health (CAMHS) Nurse	Diploma level nurse with specialty training in CAMHS to lead a new service supporting vulnerable children and adolescents with mental health problems in Kisiizi and its community. This expertise will significantly improve the quality of services offered to this vulnerable group and also allow training of other members of staff.	10m (£2,040)
Lab Scientist	Increased Quality level, potential for Lab Assistant training course	30m (£6,122)
Human resource co-ordinator	Good HR systems and staff support, reduce pressure on Administrator and MS. Previous post not replaced due to cash flow challenges with impact of lockdown due to pandemic.	30m (£6,122)
Second specialist surgeon full time x 1	We currently have one surgeon so there is a gap when he is away for general and orthopaedic specialist expertise. A second surgeon will give better cover and training of our interns and medical officers and allow sub-specialty developments and expansion of services such as endoscopy due to start in September 2020.	70m (£14,285)
Visiting Ear Nose & Throat surgeon	Monthly ENT specialist clinic / surgery to support our clinical team. (currently we have no specialist service)	18m (£3,673)
Visiting Dental Surgeon	To provide specialist service and support for our dental clinical officer, initially coming once every month to run a clinic	18m (£3,673)
Occupational Therapist	To support hospital & community service including Rehabilitation, Mental Health etc.	12m (£2,449)
Diploma social worker	To support vulnerable patients including mental health & rehab, alcohol & palliative care services etc.	8m (£1,632)

Ophthalmic Clinical Officer	To strengthen eye services from our current provision of an Ophthalmic Assistant	35m (£7,142)
	D] VEHICLE SUPPORT	
1] Ambulance replacement 4WD	Landcruiser to cope with rough murram roads. Safe movement of patients to and from Kisiizi. Current vehicle is over 15yr old. We have increasing referrals coming to Kisiizi from the region e.g. about 20 – 25 mothers a month with premature labour or complications needing expert care and we also collect sick children and transport some patients to regional centres e.g. for CT scan. Toyota quote for new fully equipped ambulance on the road includes insurance	66,000 USD (= £50,003)
2] Community Staff minibus Hiace 4 X 4, 14 seater, roofrack	To operate community clinics, staff trips to Kampala, School of Nursing activities etc. Allowing larger multi-disciplinary teams to travel so offering more services e.g. more community lab investigations. Will also support Kisiizi Falls project that in turn helps the Good Samaritan Fund of the hospital. Toyota quote on the road including taxes & insured	64,330 USD (£48,738)
3] Truck 10 ton	For movement of hospital construction materials, supplies, logistics etc. especially with the expected major project to build a new Maternity and Neonatal Unit in the next 3 years as well as the projects listed below. Will avoid the current significant costs of hiring lorries.	175m (£35,714)
4] Motor cycle	Needed for second field officer to strengthen community health insurance scheme support	4m (£816)
	E] BUILDING / INFRASTRUCTURE SUPPORT	
Surgical Ward extension & renovation & roof repair	Ward is situated in the original flax factory building constructed in 1944. The floor needs to be replaced with terrazzo for better infection control. The extension will provide more space and improved patient accommodation. The main roof is sound but the roof of a previous extension is potentially dangerous and needs to be replaced in conjunction with the new extension. The changes will also allow needed expansion of the chapel.	186m (£37,959)
Mortuary	Current mortuary is old and needs rebuilding to provide appropriate facility for relatives to visit etc. (Move container to allow easy vehicle access.)	42m (£8,571)
Rehabilitation Unit	New floor (terrazzo)	29.4m (£6,000)
Maternity toilets	Inadequate current provision for patients and staff in existing unit	12m (£2,449)
Main Entrance complex	Initial handwash, first triage and security, welcome / reception desk / insurance scheme office/ records office patient information point, issuing of wristbands for patients and attendant passes etc.	100m (£20,408)
Laboratory extension terrazzo floors, aircon. Etc.	Current lab is overcrowded with inadequate workspace. If the Insurance Scheme is moved to the Main Entrance complex as above then the Lab can take over that adjoining area so only needs furniture and terrazzo putting in.	15m (£3,061)
Accommodation student nurses 180 Students	2 – storey hostel situated behind current hostel will bring students closer to the School improving sense of community.	856m (£174,693)
Accommodation staff x 16 units (2x8)	Currently we have staff living outside our compound. If we can accommodate them it will facilitate on-call work, allows electricity support, reduces accommodation allowances costs paid by hospital.	320m (£65,306)
Upgrade 2 rooms lower med	Install en-suite facilities to two rooms in lower medical ward block for isolation cases or private patients	9m (£1,836)
Medical Specialist house	Retention of senior specialist staff includes satisfactory accommodation.	174m (£35,510)
New clinic complex	Ophthalmology, ENT, Dental clinics; ? separate adolescent facilities	380m (£77,551)
School of Nursing main	Large Multi-function for plenary lectures, examinations, assemblies , graduations etc. (around 300-350 students in school)	547m

hall		(£111,632)
NEW WARD BLOCK (in next 3-4 years)	Multi-storey purpose built high quality Maternity & Neonatal Unit. Current facilities over-crowded due to demand. Initial feedback from UK architect awaited. Links to plans for an adjoining Intensive Care Unit. Will need fully equipping.	(see notes)
Building for CT scanner		42m (£8,571)

	F] Miscellaneous	
Radio show publicity for Community Health Insurance Scheme	Provision of financial protection and improved health to more communities. Larger membership improves the sustainability of the Scheme by spreading risk and economy of scale. Supports the hospital by increased patient numbers. More opportunities for health promotion and screening to prevent avoidable disease.	Radio 10 hours over 4 months = 6m (£1,224)
Subsidy support for key patient services	Treatment of women found to have operable carcinoma of the cervix (estimated needs 12 hysterectomies a year) who may die without treatment but cannot afford the surgery.	1.2m x 12 = 14.4m (£2,938)

APPENDICES

Appendix 1 Operating Theatre instrument sets& Sterilisation Drums

Appendix 2 Instruments for Surgical Ward & Sterilisation Drums

Updated 19-08-20

Notes for future plans not included in above list:

** New Maternity and Neonatal Unit*

This is a big project needed because demand for maternity services outstrips our facilities. For example our Special Care Baby Unit currently has some incubators in what is really a corridor area. A UK Architect visited last year and is working produced some option plans concept. The site is that of the old mental health building now being used as an Isolation Unit for any Covid-19 suspects. It is anticipated that the new block will be a 2-storey high quality construct with excellent facilities for both patients and staff.

We also plan to have a 4-bed intensive care unit (ICU)to provide ventilator support and peritoneal dialysis (a process that supports kidney function for a few days while patients recover from acute kidney injury). Work on detailed departmental plans is ongoing but we would aim to start the project in 2022.

We anticipate the Maternity / Neonatal Unit will cost over £600,000,000 but detailed estimates obviously await the plans and then quantity survey reports.

Appendix 1

STATE OF INSTRUMENT				
Following the requested list of the needed instruments				
AVAILABLE in the JMS catalogue				
Caesarean set (INSSET5008)	2,043,786/=-	4	8,175,144	/=-
Dilatation and Curettage (ERPC) set (INSSET5003)	1,495,453/=-	3	4,486,359/=-	
Hernia and Appendectomy set (INSSET5009)	1,420,297/=-	3	4,260,891/=-	
Minor Surgery set=STS Set (INSSET5010)	1,607,910/=-	3	4,823,730/=-	
			21,745,724	
The rest of the other sets are not available on the JMS catalogue. I therefore suggest we purchase the following instruments to improve on the sets we have;				
Dissecting forceps, straight, 130mm NT (INSFORC015)	5,253/=-	5	26,265	
Dissecting forceps, straight, 150mm, MACINDOE (INSFORC034)	11,534/=-	5	57,670	
Dissecting forceps, straight, NT, 155mm, GILLES (INSFORC031)	11,896/=-	5	59,480	
Dissecting forceps, straight, T 150mm, GILLES (INSFORC032)	13,440/=-	5	67,200	
Dissecting forceps, straight, NT 200mm, (INSFORC016)	10,403/=-	5	52,015	
Grasping tissue forceps, 100mm, Babcock (INSFORC049)	22,614/=-	10	226,140	
Grasping tissue forceps, 205mm, Babcock (INSFORC047)	29,610/=-	10	296,100	
Tissue forceps, straight 145mm (INSFORC004)	17,274/=-	10	172,740	
Tissue forceps, straight 200mm (INSFORC028)	11,550/=-	20	231,000	
Tissue forceps, 150mm, Allis (INSFORC044)	17,274/=-	20	345,480	
Caesarean haemostatic forceps, 220mm, Green Armytage (INSFORC039)	29,809/=-	16	476,944	
Artery forceps, straight, 160mm, KOCHER (INSFORC011)	13,459/=-	10	134,590	
Artery forceps, CURVED, 150-160mm, KOCHER (INSFORC010)	11,928/=-	10	119,280	
Needle Holder, 140mm, Mathieu (INSGEN007)	24,326/=-	5	121,630	
Needle Holder, 170mm, Mathieu (INSGEN007)	31,554/=-	5	157,770	
Needle Holder 150-160mm, Mayo-Hager (INSGEN009)	15,060/=-	6	90,360	
Needle Holder 180mm, Mayo-Hager (INSGEN010)	19,719/=-	6	118,314	
Steriliser drum, 3900IA by 290mm (SPASTER034)	331,514/=-	2	663,028	
Steriliser drum, 3900IA by 140mm (SPASTER033)	744,536/=-	2	1,489,072	

26,551,202

THRU/ THE OFFICE OF THE PNG
KISISI HOSPITAL

SURGICAL WARD REQUISITION

ITEM	QUANTITY	COST	TOTAL
1. Wound Dressing Sets	5	569143	2845715
2. Emergency Tray (Box)	1	401251	401251
3. Wound Dressing Trolleys	2	642510	1285020
4. Drums	2 big	630249	1260498
5. Drums	2 medium	487708	975416
6. Kidney Dishes, Size 20.7cm, 24.7cm	10, 10	10043, 15844	258870
7. HDU Beds	3	1097957	3293871
8. Otoscope (1000 pcs)	1	151629	151629
9. Ear Syringe (Metal)	1	63833	63833
			8,300,189

Prepared by,

SAMSON MUJUNI
SURGICAL WARD
IN-CHARGE

Updated Feb 2021